Holographic Bird Deterient

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1 Project Overview

This project aims to design and implement an optical holographic bird deterrent using pre-processed video to simulate realistic behaviors of predatory birds, specifically focusing on activities such as preening to maximize deterrent effectiveness. Unlike static visual deterrents, dynamic optical holograms significantly reduce bird habituation due to their lifelike and unpredictable appearances. The device will feature weather-tolerant construction suitable for extended outdoor deployment, utilizing durable optical components and protective enclosures to ensure reliable operation under various environmental conditions. Additionally, the project will include the development and proposal of an experimental design to rigorously assess the effectiveness of the holographic deterrent, comparing bird visitation rates between areas employing the holographic display and control sites without the device.

2. Design Process

The design process began with identifying the core requirements for an effective bird deterrent: lifelike motion, outdoor durability, and easy visibility. Early brainstorming considered various methods such as laser projection and volumetric displays. The team ultimately selected a holographic-style optical projection approach due to its proven ability to mimic motion in a compact footprint, and its easy constructibility / reproducibility.

To start our design we split the system into four major parts: lens assembly, compute module, video acquisition, and video processing. This allowed us to better divide the work up and keep us on track to finish by the end of the semester. Collaboration between team members ensured integration between hardware design, video data preparation, and different testing strategies.

2.1 Lens

The lens of the project is used to give the “3D” effect to the projection, adding realism that will hopefully help deter birds. There are a couple different ways that you can use to project a realistic looking object. There is a flat sheet lens, prism lens, or a rotating lens mainly used in volumetric displays. To adhere to our goal of having 4 directional viewing and a simple concept we went with the prism which is made up of 4 trapezoid sheets angled and connected together to give the 3D effect. The sheets were made up of plexiglass, as it provides the best reflection for how durable it is.

To get the correct dimensions we used specific dimension calculations based on our monitor size we are using. The width of the monitor was 15 inches so the top length of the pieces will be 15 inches. To get the other dimensions you take the width / 6 = 2.5 to get the bottom edge of the piece. Then multiplying that value of 2.5 in this case by 3.5 will give us the height value of 8.75 inches that we need. Once we had our correct dimensions we traced out the pieces and cut them using a table saw. To keep the shape of the final design as accurate as possible we sanded down the edges of the pieces to keep them as aligned as possible so our projection will turn out as clean as we can make it. To assemble the lens you just have to connect the edges of the pieces together to form a prism looking shape. To do this we decided to use tape as a guide and then connect them together with a hot glue gun, to keep the clearness of the prism while providing adequate durability. This can be scaled to any size we need depending on the size of the monitor. For future semesters we will try to increase the size of the monitor and lens to get a more accurate size projection.

Figure 1. *Image of the outlined pieces for cutting*

Figure 2. *Image of the cut pieces lined up with tape to be assembled*

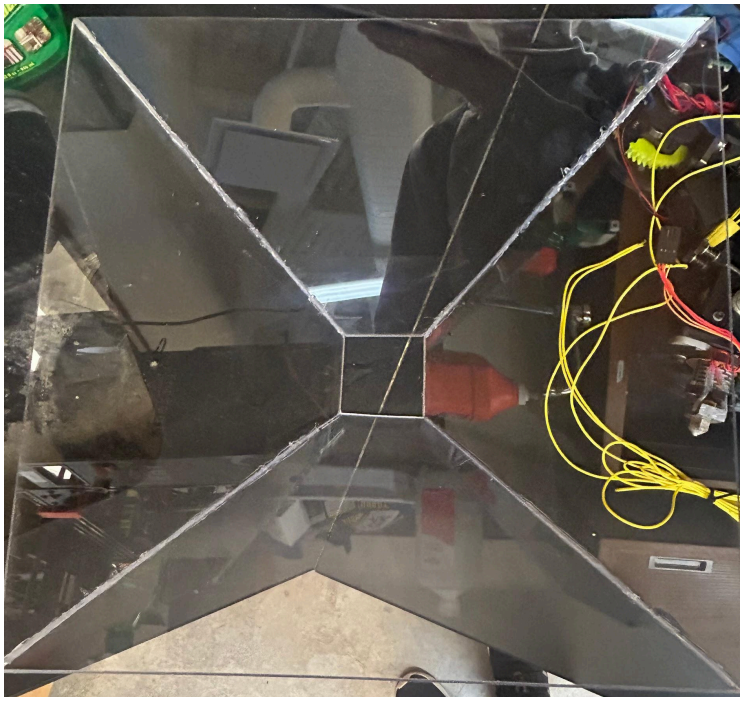

Figure 3. *Image of the final lens product glued together*

2.2 Compute Module

We used a Raspberry Pi 5 as the primary computer. This had the latest version of Raspberry Pi 5 OS installed onto the SD Card. A USB drive was formatted with FAT 32 for windows compatibility and this is where the video files were stored. FAT 32 may not be adequate for large files (larger than 4 GiB) due to file system limitations, if the user wanted to supply an especially long or high quality file they would need to format it with a native Linux file system such as ext4. The system was configured to replace TTY 1 with a wayland session which plays the bird video. This was done using greetd, which was configured to run a custom bash script after boot. The video is run by the MPV video player inside of the Cage wayland compositor. MPV was configured to accept playback controls using a UNIX socket for integration with IO systems. All programs were acquired directly from the APT package manager.

IO was handled using short c++ programs which interfaced with the Raspbury Pi GPIO pins. Input listens on a user supplied pin for any signals, if it detects a signal it sends a JSON command to the MPD unix socket telling it to either swap the bird video with a black screen or vise-versa. Output is managed by a custom text format which tells the device to enable a given pin at a given time. Currently, this is not synchronized with the video output. The IO programs are started by systemd during startup. They are intentionally started fairly late during the startup sequence to hopefully align with the startup of the video playback, however the startup order of systemd services is non-deterministic since system updates may add or remove services. Finally, in the future the code should be modified to support standard protocols such as I2C for better integration with other devices.

2.3 Video Acquisition

The deterrent's effectiveness relies on the realism of the projected content. Our video acquisition was focused strictly on sourcing high-definition, high-frame-rate footage of predatory birds. In our case the final goal is a Merlin which will be used to deter migratory birds. To get a first test going we used what we had available to video a small bird that was caught. We wanted to have all steps of the project be as open source and easy to do as possible, so to record we used a phone recording normal and slo-mo clips that we can use to edit and project. The phone used was able to record at 4k 60fps for normal videos, and 1080p 120fps for slo-mos. This was the simplest and best option available since it was easy to use and helped us meet our 120fps goal. We plan on working with Professor Wolfe and contacting other people to try and get a high quality video of a Merlin for next semester.

Figure 4. *Image from one of the acquired videos showing the bird and how it was captured*

2.4 Video Processing

Kdenlive was used to process the video files. A selection of video was identified as looping to a satisfactory degree. This section was cut out of the video. An object mask of the bird was created using the SAM 2 machine learning model inside of Kdenlive. After this step, the project file was somewhat unstable, so a full quality render was done. This video file was then imported into another project file. Four copies of this file were added to the timeline. Each instance had a color mask set to black, since transparent pixels were exported as black due to the limitations of the video file format. Each instance of the video was then given a transform effect with values that depend on the aspect ratio and the size of the video to correctly align the birds for presentation. Since alignment with the lens was determined to not be particularly precise, trial and error was used to select correct values.

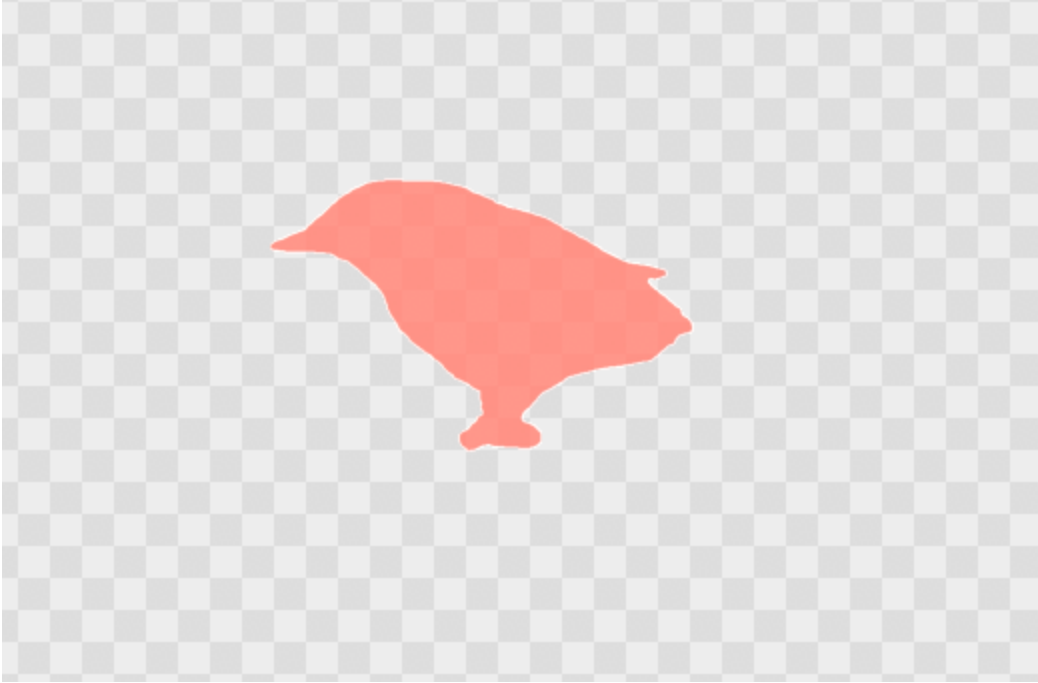

Figure 5. *Still frame of the alpha mask. Generated by SAM 2*

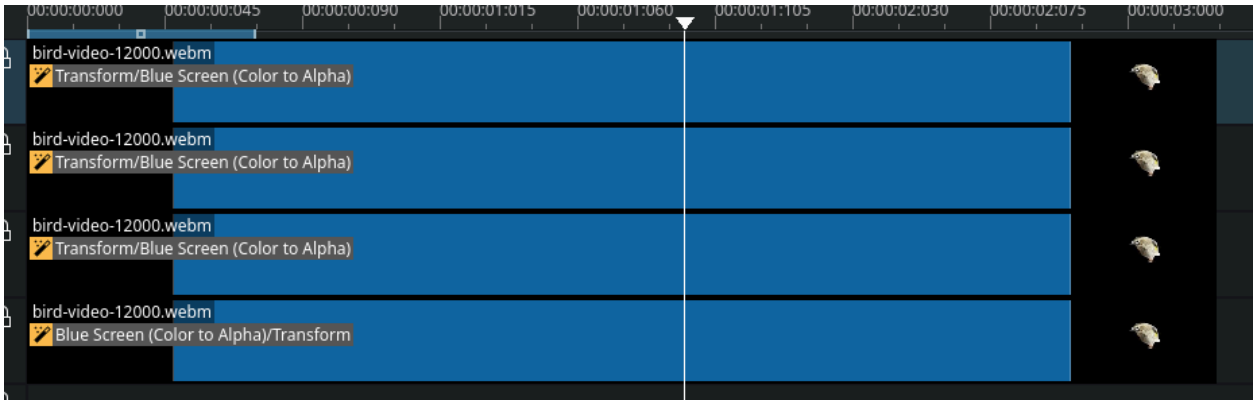

Figure 6, *Video timeline of the final project. Contains four instances of the masked video with transform and bluescreen effects.*

3 Bill Of Materials

Bill Of Materials for OSHE - Holobird				
Item	**Part Number/Link**	**Description**	**Amount**	**Price**
Raspberry Pi 5	2648-SC1431-ND Link	Low cost computer used to display the videos of the birds for the hologram	1	$64.26
Plexiglass	N/A - Ace Hardware	The material selected to reflect the video of the bird from the monitor	1 30” by 30” Sheet	$38.16
Monitor	OSHE Lab Monitor	Monitor from OSHE Lab	1	$0.00
Heat Sink	RSP-HS-PI4	Raspberry Pi 4 Heatsink. From OSHE Lab.	1	$4.95
Flash Drive	SDCZ50-008G-B35 Link	Flash drive used in order to store the video file to play on the Pi. From OSHE Lab	1	$9.49
8 Gb SD Card	SDSDQ-8192 Link	SD Card used for booting the Pi. From OSHE Lab.	1	$9.21
Totals			6	$126.07

Tools Used for OSHE - Holobird				
Tool	**Part/Acquisition**	**Description**	**Amount**	**Cost**
Table Saw	Makers Space	Used to cut the plexiglass to proper dimensions	1	$0.00
Phone	Members Phone	Used to record the video of the bird that is projected	1	$0.00
Glue Gun	OSHE Lab	Used to glue together the plexiglass pieces	1	$0.00
Kdenlive	Github	Open source video software used to edit the bird video	N/A	$0.00
SAM 2	Github	Object detection algorithm used to create	N/A	$0.00

Tools Used for OSHE - Holobird				
		alpha masks		
SD Card Reader	ODK550 Link	Used to read they SD card with the Raspberry Pi, From OSHE Lab	1	$12.99
greetd	SourceHut	Session manager	N/A	$0.00
MPV	Website	Embeadable video player	N/A	$0.00
Cage	Github	“Kiosk style” wayland compositor.	N/A	$0.00
libgpiod	kernel.org	GPIO controller daemon	N/A	$0.00
Totals				$12.99

4 Functional Testing

Since the outdoor casing is yet to be developed, all functionality tests had to be performed indoors. The display system was tested by building a “mock up” of the final device using the monitor in the OSHE lab combined with the Lens. Input testing was done by manually emulating the protocol using a wire to short pins on the Raspberry Pi. The output was tested using a Fluke Digital Multimeter and watching the voltage levels.

Figure 7. *Still frame of the device working using a test video.*

5 Bird Vision Requirements

Bird vision played an important role in determining the visual requirements for our holographic deterrent. Many bird species process motion much faster than humans, meaning the video needed a high enough frame rate to appear smooth and lifelike to the birds. For most birds you would need about 120 fps to avoid choppy movement that birds might ignore. Birds also have a wider and more complex color perception range, including sensitivity into the ultraviolet spectrum, so we focused on looking into high-contrast footage to ensure the projection stands out even without UV reproduction if necessary. Movement patterns were another important factor, natural behaviors like preening or head-turning are more believable to birds than aggressive poses held statically for long periods. Finally, brightness had to be considered because outdoor environments often wash out dim displays. Ensuring the projected image is bright, sharp, and visible from multiple angles helps make the deterrent more noticeable and reduces the chance of habituation.

6 References

Potier, Simon, et al. "How Fast Can Raptors See?" *Journal of Experimental Biology*, vol. 223, no. 1, 1 Jan. 2020, https://doi.org/10.1242/jeb.209031 Accessed 5 Dec. 2025.

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